Automated Flashlamp Microscopic Flaw Inspection

Sponsors: Bruno Le Galloudec, Evan Carroll
Engineers: Lucas McLaren, Benjamin Nelson, Samantha Paul, Ethan Waterman
Advisor: Dr. Dennis O'Connor

MMEM Capstone Design Program

Project Overview

The National Ignition Facility (NIF) performs experiments in high energy fusion research. Thousands of flashlamps are inspected manually before use, taking hours for each flashlamp. The goal of this project is to automate the process of flashlamp inspection while maintaining the quality of inspection.

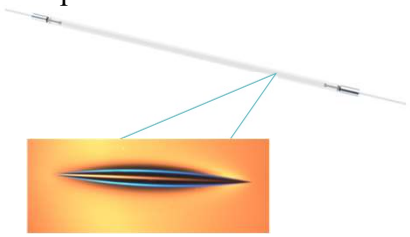


Fig 1. Flashlamp defect.

Requirements

- Image entire flashlamp
- Identify defects >0.3*mm*
- Categorize and store defect data
- Cycle time of 30*mins*
- Accommodate flashlamp and tube.

Design Solution

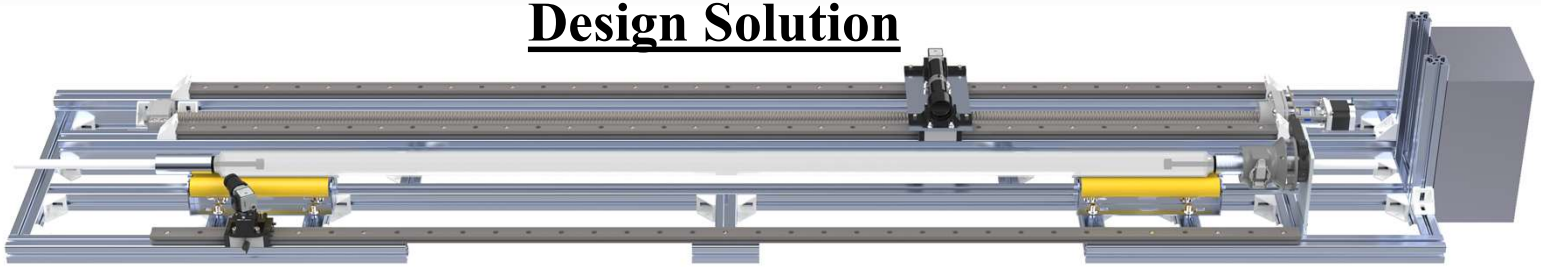


Fig 2. Full system assembly.

Mechanical: The flaw inspection system employs two mechanical subsystems to achieve full imaging capabilities.

- **Linear Translation**
  - 2.1m Thomson Linear Ball screw
- **Rotational Translation**
  - Thorlabs Rotation Stage
  - Custom 3D printed latching clamp



Fig 3. Latching clamp.

Optical: The optical system uses two cameras to image the entire surface of the flashlamp and the electrode end caps.

- **Primary Camera**
  - 9.1 Megapixel Resolution
  - Telecentric Lens
- **Secondary Camera**
  - 5 Megapixel Resolution
  - Fixed Focal Length Lens



Fig 4. Primary and Secondary Cameras.

Image Analysis: Images taken by the system are filtered using OpenCV machine learning, coded in Python. Previously imaged defects, as well as future images, will be used to train and validate the models which analyze these images. Types of images are airlines, bubbles, scratches, etc.

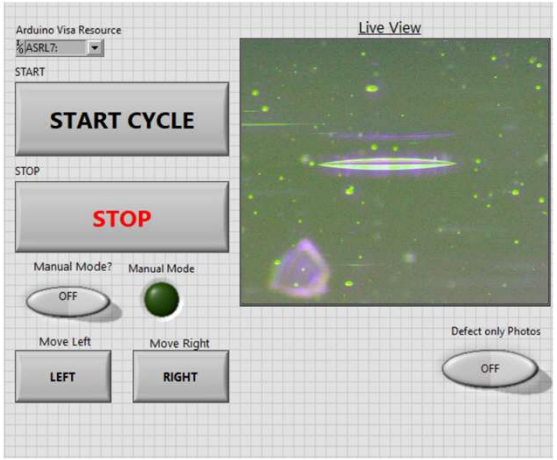


Fig 5. LabVIEW GUI.